CANTEEN / FOOD ORDERING SYSTEM

PROGRAMMING WITH PYTHON

(202044504)

A Mini Project Report submitted in partial fulfilment of the requirements for the Degree of

Bachelor of Technology in Information Technology

Semester V

SUBMITTED BY

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CERTIFICATE

This is to certify that **Mr. Ansh Vinod Patel**, a student of the **Bachelor of Technology (Information Technology)** programme, **Semester V**, has successfully completed the mini project entitled **"Canteen Food Ordering System"** for the course **"Programming with Python" (Course Code: 202044504)** under the guidance of **Mr. Nayan Mali**.

This project demonstrates the practical application of Python programming concepts, Object-Oriented Programming (OOP), dictionaries, Tkinter graphical user interface (GUI), and MySQL database connectivity. The project report is submitted in partial fulfilment of the academic requirements of the **Professional Core Course**.

We appreciate the efforts made by the student in completing this mini project.

Date: _____

Place: _____

Course Faculty
Mr. Nayan Mali
Signature: _____

1. Abstract

The Canteen / Food Ordering System is a desktop application developed using Python, Tkinter and MySQL. It helps a college canteen manage food items, accept customer orders, calculate bills and keep an order history. The graphical interface allows the user to select menu items, add them to a cart, adjust quantities, choose a payment method and place an order. The order and its individual items are stored in a relational database. The project demonstrates Object-Oriented Programming, dictionaries, GUI programming, database connectivity, SQL operations and exception handling.

2. Introduction

A college canteen serves many students and staff members during short breaks. Recording orders and calculating totals manually can take time and may cause mistakes. A computerized ordering application can make the process more organized. This project provides a simple interface for selecting available food, calculating the payable amount and recording completed orders. It is designed as an educational prototype and can be extended with login accounts, stock control, online payments and kitchen order status.

3. Problem Statement

Manual canteen ordering may involve handwritten notes, repeated price calculations and difficulty finding previous transactions. The project addresses these problems by providing a digital menu, cart-based ordering, automatic bill calculation and a searchable recent-order view. It also stores order details in MySQL so information remains available after the application closes.

4. Objectives

- Display menu items with their categories and prices.
- Allow users to add, remove and change the quantity of items.
- Calculate the total bill automatically.
- Store customer, payment, order and order-item information in MySQL.
- Generate a bill showing the order number and item-wise totals.
- Allow menu items to be added or updated through the GUI.
- Demonstrate OOP, dictionaries, Tkinter and database operations.

5. Scope

The application is intended for a small college canteen or as a classroom demonstration. It supports one local application user at a time, sample menu data, basic payment-method recording and recent-order viewing. It does not process real card or UPI payments; the payment option is recorded for demonstration only. It does not currently include authentication, inventory deductions, tax configuration, refunds or multi-terminal synchronization.

6. Hardware and Software Requirements

- Hardware: desktop or laptop, keyboard, mouse and sufficient memory to run Python and MySQL.

- Operating system: Windows, macOS or Linux with a desktop environment.
- Language: Python 3.
- GUI library: Tkinter and ttk.
- Database: MySQL Server.
- Connector: mysql-connector-python.
- Editor: Visual Studio Code or any Python IDE.

7. System Design

The application has three main layers. The presentation layer is the Tkinter interface containing the food menu, cart, customer form and action buttons. The application layer validates inputs, updates the cart and calculates totals. The database layer uses the CanteenDatabase class to run parameterized SQL queries and persist data. Separating the GUI from database methods makes the program easier to read and maintain.

8. Main Modules

- Menu module: retrieves available food items from the menu_items table.
- Cart module: stores selected items and quantities in a Python dictionary.
- Billing module: calculates line totals and the overall total and displays a bill.
- Order module: saves the order header and all order lines in a database transaction.
- Order history module: displays up to 100 recent orders.
- Menu maintenance module: inserts a new item or updates an item with the same ID.

9. Database Design

The database is named canteen_db and contains three tables.

Table	Purpose	Important fields
menu_items	Stores available food and price	item_id (PK), name, category, price, available
orders	Stores one record per order	order_id (PK), customer_name, payment_method, total, order_time
order_items	Stores each food line in an order	order_item_id (PK), order_id, item_id, item_name, quantity, unit_price, line_total

One order can contain multiple order_items records. The order_items.order_id foreign key refers to orders.order_id. The item name and unit price are also saved with each order line so the historical bill retains the details used at the time of purchase.

10. Object-Oriented Programming

The project defines a CanteenDatabase class for database connectivity and operations, and a CanteenApp class for the graphical interface and order workflow. Objects keep related behavior together. The main function initializes the database and starts the application. This structure is more maintainable than placing every operation in a single long function.

11. Use of Dictionaries

The DEFAULT_MENU dictionary provides sample menu records keyed by item ID. The cart dictionary uses an item ID as its key and stores the item details together with the selected quantity. Dictionaries make it convenient to find an item, update its quantity and calculate the subtotal. The MySQL database remains the persistent source for menu and order records.

12. User Interface

The main window is divided into a food menu on the left and the order cart on the right. The menu shows item ID, name, category and price. Buttons add a selected item, increase or decrease quantity, remove a line, place an order, show recent orders and add or update a menu item. Customer name and payment method are entered before checkout. A separate window displays the generated bill.

13. Algorithm

1. Start the program and connect to MySQL.
2. Create the database and tables if required; seed the sample menu if it is empty.
3. Load the available menu into the GUI.
4. When the user selects a food item, add it to the cart or increase its quantity.
5. Recalculate each line subtotal and the cart total after every change.
6. Validate the customer name and ensure the cart contains at least one item.
7. Insert the order header and its item lines in one transaction.
8. Commit the transaction and display the bill; if a database error occurs, roll back the transaction.
9. Clear the cart after a successful order and allow the user to continue.

14. Testing

Test case	Expected result	Status to verify
Launch with MySQL running	Window opens and menu loads	Run locally
Add one menu item	Item appears in cart with quantity 1	Run locally
Increase quantity	Quantity and total increase	Run locally
Decrease quantity to zero	Item is removed from cart	Run locally
Place order with empty cart	Warning message appears	Run locally
Place order without customer name	Validation warning appears	Run locally
Place a valid order	Order and order lines are saved; bill appears	Run locally
Open recent orders	Saved orders are listed	Run locally
Add/update menu item	Menu record is inserted or updated	Run locally

Testing should be performed on the student's own computer after configuring MySQL. The status column is intentionally left as a verification checklist rather than claiming tests were executed on the student's machine.

15. Exception Handling and Data Safety

The application handles invalid numeric menu input, missing customer names, empty carts and MySQL connector errors with messages. Order creation uses a database transaction: all order records are committed together, and a failure triggers rollback. SQL statements use parameter placeholders for values rather than building queries from user-entered text. The database password should remain local and should not be included in a public repository or submitted report.

16. Limitations and Future Enhancements

- Add administrator login and role-based permissions.
- Add stock quantity and automatically reduce inventory after a successful order.
- Add order status such as Pending, Preparing and Ready.
- Add date-range sales reports and daily revenue summaries.
- Add search and category filters to the menu.
- Add receipt printing and export to PDF or CSV.
- Integrate a real payment gateway only with suitable security and testing.
- Add a web or mobile interface for customer self-service.

17. Conclusion

The Canteen / Food Ordering System provides a practical example of applying Python to a familiar real-world process. It combines a Tkinter GUI, dictionary-based cart handling, OOP design and MySQL persistence. The system automates basic ordering and billing while retaining a record of completed transactions. The project also provides a foundation for future improvements such as inventory management, user authentication and analytics.

18. References

- Python documentation: https://docs.python.org/3/
- Tkinter documentation: https://docs.python.org/3/library/tkinter.html
- MySQL documentation: https://dev.mysql.com/doc/
- MySQL Connector/Python documentation: https://dev.mysql.com/doc/connector-python/en/